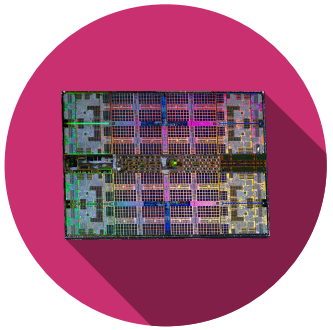
Computer Architecture

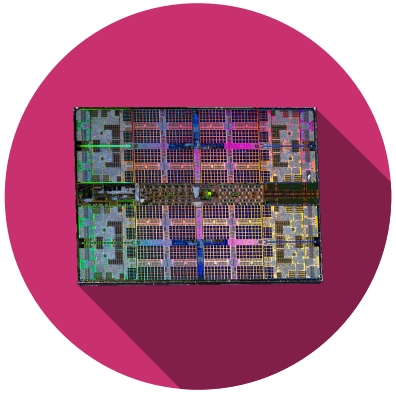
# Performance

Per

Information and Communication Systems program

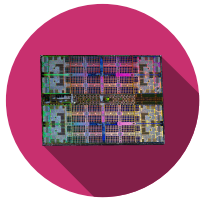


Silvan Zahno silvan.zahno@hevs.ch



# Benchmarks

# Benchmarks

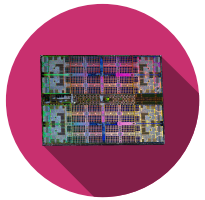


Judges the relative performance of a CPU

**Key terms**

- MIPS
- MFLOPS
- MHz
- FPS
- Render Time
- Dropped Frames

# Benchmarks

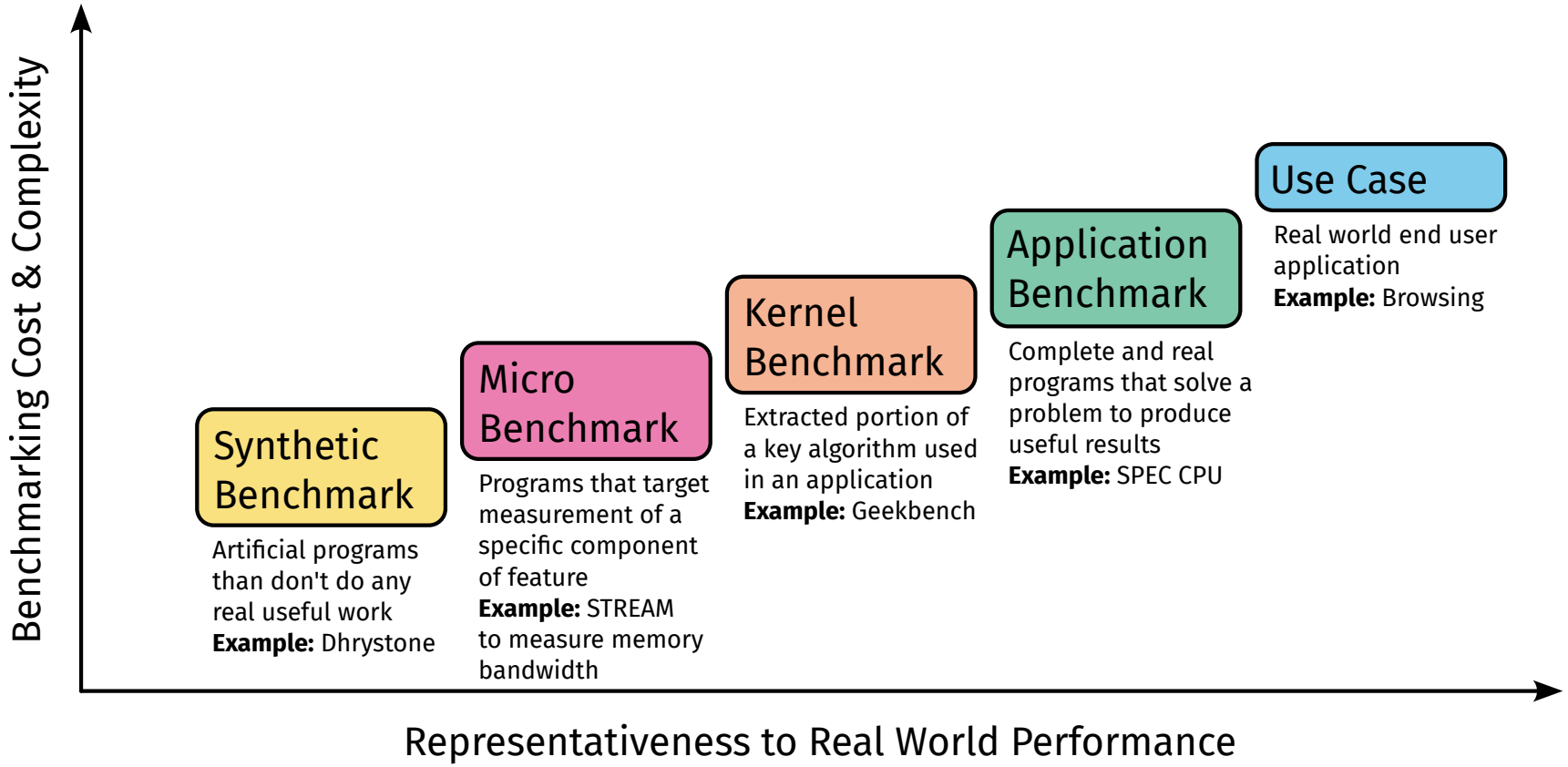


Judges the relative performance of a CPU

**Key terms**

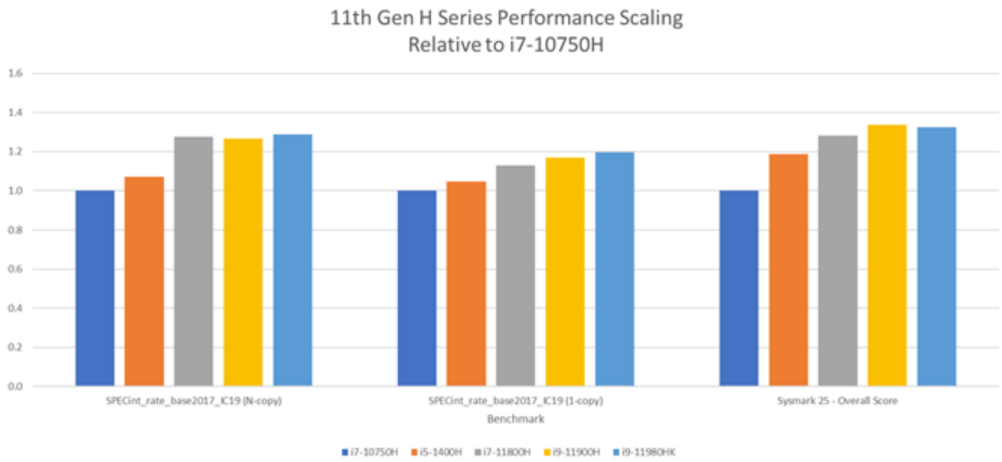
- MIPS – Million Instruction per Second
- MFLOPS – Million Floating Point Operation per Second
- MHz - Frequency
- FPS – Frame per second
- Render Time – Time to render 2D/3D scene
- Dropped Frames – number of frames lost in streaming

# Spectrum of Benchmarks



# SPEC Benchmark SPEC CPU 2017

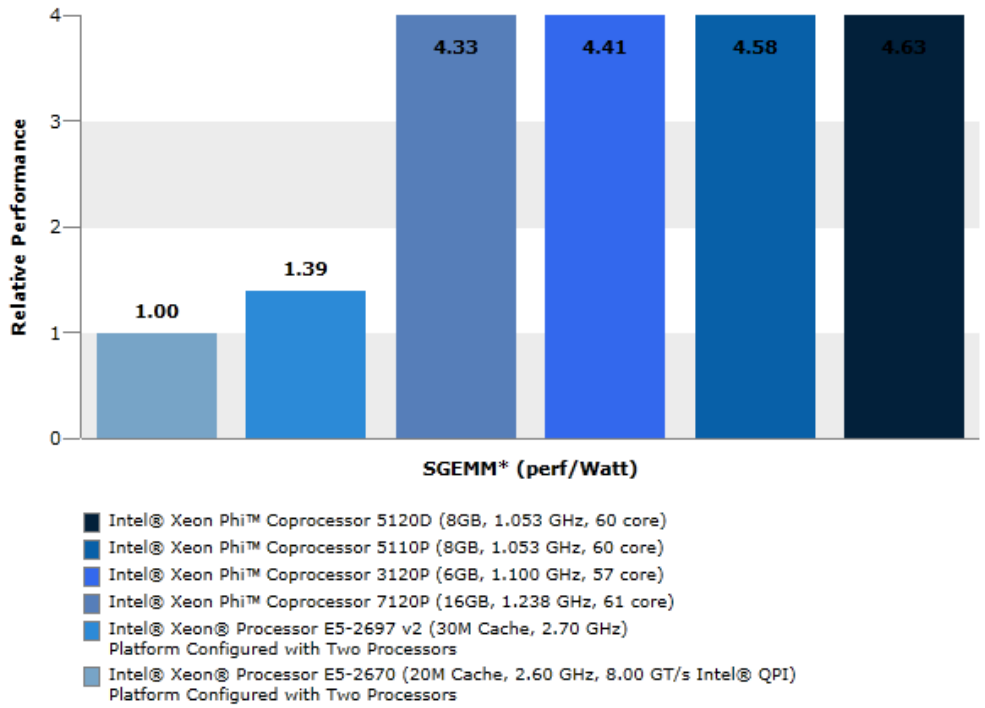
- **S**tandard **P**erformance **E**valuation **C**orporation
- Measures integer and floating point operation performance
- Contains 10 integer and 13 floating point operations
- Compute intensive, concentrates on CPU and memory



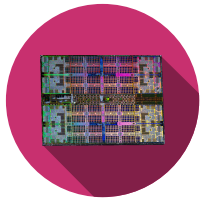
# Linpack

## Performance per Watt

- Performances of double precision vector and matrix operations
- Ofen used by HPC communicate to measure FLOPS of a processor



# Aspects of Computer Performance



Time

- Response time / latency / execution time
  - Time between start and completion of event / task / program (n seconds)

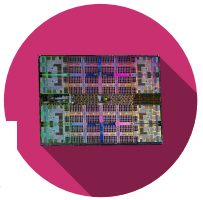
Other Aspects

- Availability, scalability, performance per watt

# Benchmark Suites

| Benchmark | Organization | Description | Comment |
|---|---|---|---|
| Dhrystone | Open Source | Synthetic benchmark measure compute-intensice integer performance | No longer representative of modern workloads |
| CoreMark | EEMBC | Popular benchmark to measurecompute-intensive integer performance | Result published through EEMBS, continuation of Dhrystone |
| SPEC CPU | SPEC OSG CPU | Application benchmarks that comprise of open-source real world applications. | Most popular for measuring CPU performamce |
| STREAM | Open Source | Suite of microbenchmarks to measure sustained memory bandwidth | Measure of memory latency and bandwidth |
| Geekbench 5 | Primate labs | Kernel benchmarks that measure CPU Integer, Floating Point and Memory performance | Used in industry, but with some limitation on the representation of real-world scenarios |
| PassMark | PassMark Software | Kernel benchmarks that measure CPU Integer, Floating Point and Memory performance Per | Used in industry, but with some limitation on the representation of real-world scenarios |

# SPEC Benchmarks



| SPECrate®2017 Integer | SPECspeed®2017 Integer | Language [1] | KLOC [2] | Application Area |
|---|---|---|---|---|
| 500.perlbench_r | 600.perlbench_s | C | 362 | Perl interpreter |
| 502.gcc_r | 602.gcc_s | C | 1,304 | GNU C compiler |
| 505.mcf_r | 605.mcf_s | C | 3 | Route planning |
| 520.omnetpp_r | 620.omnetpp_s | C++ | 134 | Discrete Event simulation - computer network |
| 523.xalancbmk_r | 623.xalancbmk_s | C++ | 520 | XML to HTML conversion via XSLT |
| 525.x264_r | 625.x264_s | C | 96 | Video compression |
| 531.deepsjeng_r | 631.deepsjeng_s | C++ | 10 | Artificial Intelligence: alpha-beta tree search (Chess) |
| 541.leela_r | 641.leela_s | C++ | 21 | Artificial Intelligence: Monte Carlo tree search (Go) |
| 548.exchange2_r | 648.exchange2_s | Fortran | 1 | Artificial Intelligence: recursive solution generator (Sudoku) |
| 557.xz_r | 657.xz_s | C | 33 | General data compression |

| SPECrate®2017 Floating Point | SPECspeed®2017 Floating Point | Language [1] | KLOC [2] | Application Area |
|---|---|---|---|---|
| 503.bwaves_r | 603.bwaves_s | Fortran | 1 | Explosion modeling |
| 507.cactuBSSN_r | 607.cactuBSSN_s | C++, C, Fortran | 257 | Physics: relativity |
| 508.namd_r | | C++ | 8 | Molecular dynamics |
| 510.parest_r | | C++ | 427 | Biomedical imaging: optical tomography with finite elements |
| 511.povray_r | | C++, C | 170 | Ray tracing |
| 519.lbm_r | 619.lbm_s | C | 1 | Fluid dynamics |
| 521.wrf_r | 621.wrf_s | Fortran, C | 991 | Weather forecasting |
| 526.blender_r | | C++, C | 1,577 | 3D rendering and animation |
| 527.cam4_r | 627.cam4_s | Fortran, C | 407 | Atmosphere modeling |
| | 628.pop2_s | Fortran, C | 338 | Wide-scale ocean modeling (climate level) |
| 538.imagick_r | 638.imagick_s | C | 259 | Image manipulation |
| 544.nab_r | 644.nab_s | C | 24 | Molecular dynamics |
| 549.fotonik3d_r | 649.fotonik3d_s | Fortran | 14 | Computational Electromagnetics |
| 554.roms_r | 654.roms_s | Fortran | 210 | Regional ocean modeling |

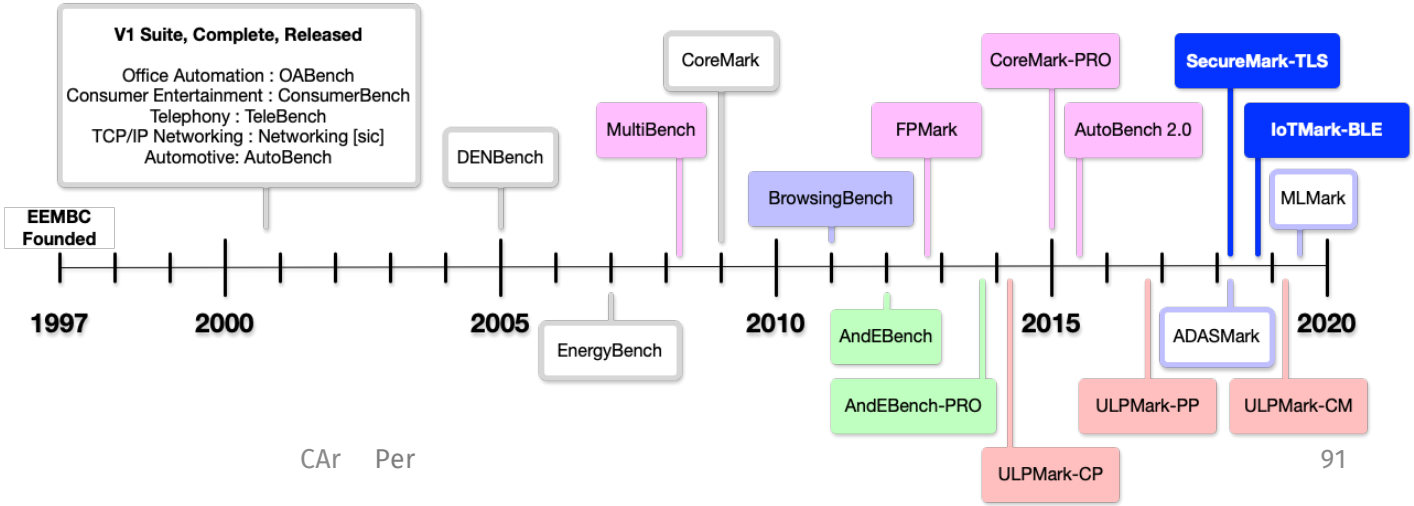
[1] For multi-language benchmarks, the first one listed determines library and link options (details)

[2] KLOC = line count (including comments/whitespace) for source files used in a build / 1000

# EEMBC Benchmarks

Benchmarks can be grouped into the following categories:

- Ultra-Low Power and Internet of Things
- Heterogeneous Compute
- Single-core Processor Performance
- Multi-core Processor Performance
- Phone and Tablet



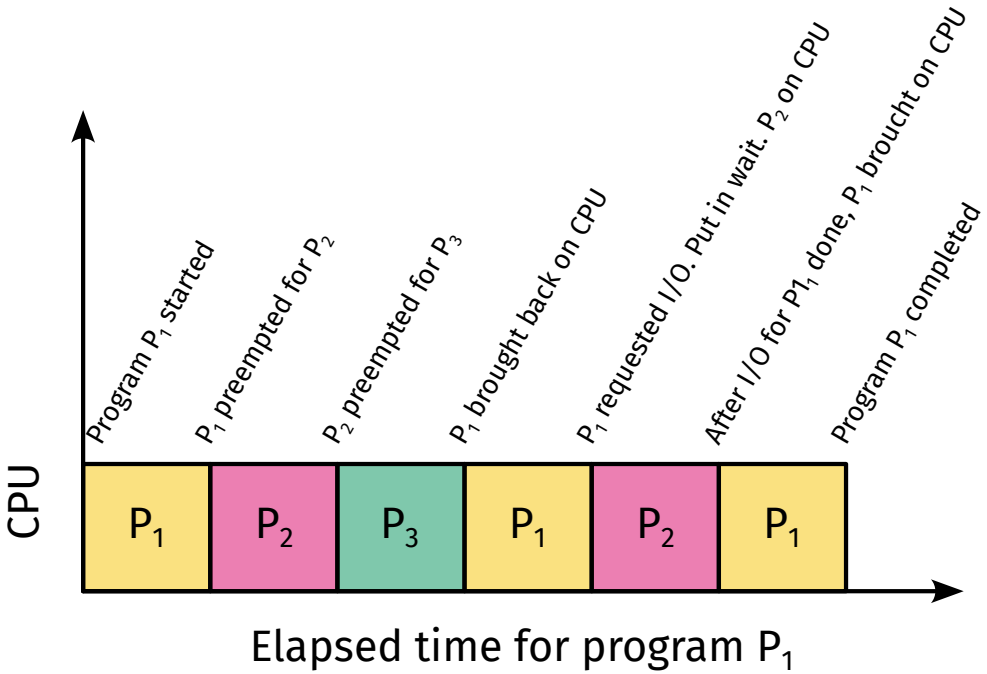
# How to Measure Execution time?

## Wall-clock time

- Elapsed time
  - Disk Access, I/O, OS overhead, ...

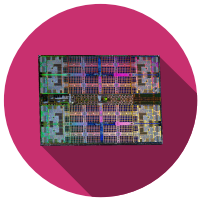
## Multiprogrammed CPI works on other process when current process stalled for I/O

## CPU-time = time CPU is computing



# Benchmark

## Exercise

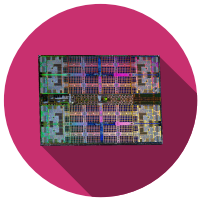


Which of the following statements are correct?

1. The wall clock time is the total elapsed time , including I/O, Operating system overhead etc.?
2. Multi-threading improves the throughput of a process
3. The CPU time does not include the I/O time
4. Multi-threading improves the execution time of a process

## Benchmark

### Exercise



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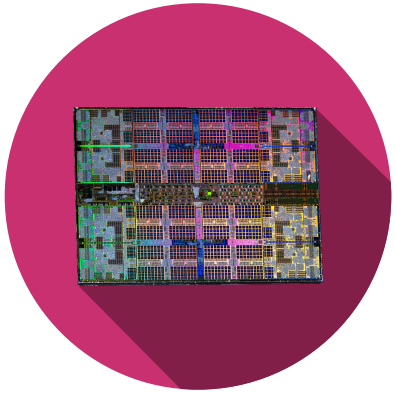


# Performance

# Summarizing Performance

- With 1 program it is clear which computer is faster
- With >1 programs, it depends

How do you aggregate their performance?

# Summarizing Performance

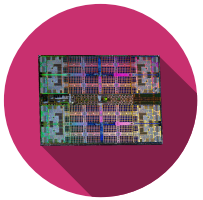
- With 1 program it is clear which computer is faster
- With >1 programs, it depends

How do you aggregate their performance?

$$ArithmeticMean = AM = \frac{1}{n}\sum_{i=1}^{n} Time_i$$

# Summarizing Performance

## Exercise

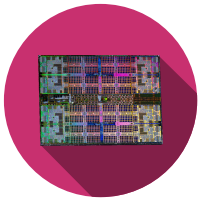


| | CPU A | CPU B | CPU C |
|---|---|---|---|
| Program $P_1$ (sec) | 1 | 10 | 20 |
| Program $P_2$ (sec) | 1000 | 100 | 20 |

1. Which CPU is the fastest for $P_1$ ?
2. Which CPU is the fastest for $P_2$ ?
3. Which CPU is the fastest?

# Summarizing Performance

## Exercise

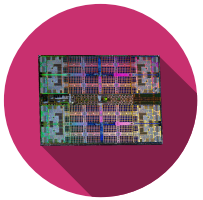


| | CPU A | CPU B | CPU C |
|---|---|---|---|
| Program $P_1$ (sec) | 1 | 10 | 20 |
| Program $P_2$ (sec) | 1000 | 100 | 20 |
| Total (sec) | | | |
| Arithmetic Mean | | | |

1. Which CPU is the fastest for $P_1$ ? => CPU A
2. Which CPU is the fastest for $P_2$ ? => CPU C
3. Which CPU is the fastest ?

# Summarizing Performance

## Exercise



| | CPU A | CPU B | CPU C |
|---|---|---|---|
| Program $P_1$ (sec) | 1 | 10 | 20 |
| Program $P_2$ (sec) | 1000 | 100 | 20 |
| Total (sec) | 1001 | 110 | 40 |
| Arithmetic Mean | 500.5 | 55 | 20 |

1. Which CPU is the fastest for $P_1$ ? => CPU A
2. Which CPU is the fastest for $P_2$ ? => CPU C
3. Which CPU is the fastest ? => CPU C

Only if the program are run equally often!

# Summarizing Performance

- What if the program not run equally often?
- Two approaches:
  - Weighted execution time
  - Normalize execution times to a reference machine and take the average

# Summarizing Performance

## Weighted Execution Time

- Program $P_i$ has a weight $Weight_i$
  - Indicates execution frequency
  - weights add up to 1
- Program $P_i$ takes time $Time_i$
- Weighted arithmetic mean:

$$WeightedArithmeticMean = WAM = \sum_{i=1}^{n} Weight_i * Time_i$$

# Summarizing Performance

## Weighted Execution Time - Example

| | CPU A | CPU B | CPU C |
|---|---|---|---|
| Program $P_1$ (sec) | 1 | 10 | 20 |
| Program $P_2$ (sec) | 1000 | 100 | 20 |
| Arithmetic mean | 500.5 | 55 | 20 |

1. Which CPU is the fastest with $w_1$=0.8 and $w_2$ = 0.2
2. Which CPU is the fastest with $w_1$=0.9 and $w_2$ = 0.1

# Summarizing Performance

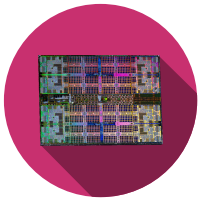
## Weighted Execution Time - Example

| | CPU A | CPU B | CPU C |
|---|---|---|---|
| Program $P_1$ (sec) | 1 | 10 | 20 |
| Program $P_2$ (sec) | 1000 | 100 | 20 |
| Arithmetic mean | 500.5 | 55 | 20 |
| Weighted mean ($w_1$=0.8, $w_2$=0.2) | 200.8 | 28 | 20 |
| Weighted mean ($w_1$=0.9, $w_2$=0.1) | 100.9 | 19 | 20 |

1. Which CPU is the fastest with $w_1$=0.8 and $w_2$ = 0.2
   - CPU C
2. Which CPU is the fastest with $w_1$=0.9 and $w_2$ = 0.1
   - CPU B

# Summarizing Performance

## Weighted Execution Time - Exercise



We want to buy a new computer. It will mostly run programs $P_1$ and $P_2$.

1. When is CPU A the best buy?
2. When is CPU B the best buy?
3. When is CPU C the best buy?

| | CPU A | CPU B | CPU C |
|---|---|---|---|
| Program $P_1$ (sec) | 1 | 10 | 100 |
| Program $P_2$ (sec) | 100 | 10 | 1 |

# Summarizing Performance

## Normalized Execution Time

- Normalize the execution time to a reference computer

$$ExecutionTimeRatio = \frac{ExecutionTime_{reference}}{ExecutionTime_{new}}$$

- Used by SPEC Benchmarks, called SPECRatio

# Summarizing Performance

## Normalized Execution Time - Example

- Do NOT use arithmetic mean to average normalized execution times!!

| | CPU A | CPU B | CPU C |
|---|---|---|---|
| Program $P_1$ (sec) | 1 | 10 | 20 |
| Program $P_2$ (sec) | 1000 | 100 | 20 |

| | Normalized to A | | | Normalized to B | | |
|---|---|---|---|---|---|---|
| | CPU A | CPU B | CPU C | CPU A | CPU B | CPU C |
| ETR $P_1$ | 1 | 0.1 | 0.05 | 10 | 1 | 0.5 |
| ETR $P_2$ | 1 | 10 | 50 | 0.1 | 1 | 5.0 |
| Arithmetic mean | 1 | 5.05 | **25.025** | **5.05** | 1 | 2.75 |

# Summarizing Performance

## Normalized Execution Time - Example

- Do NOT use arithmetic mean to average normalized execution times!!

| | CPU A | CPU B | CPU C |
|---|---|---|---|
| Program $P_1$ (sec) | 1 | 10 | 20 |
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| | Normalized to A | | | Normalized to B | | |
|---|---|---|---|---|---|---|
| | CPU A | CPU B | CPU C | CPU A | CPU B | CPU C |
| ETR $P_1$ | 1 | 0.1 | 0.05 | 10 | 1 | 0.5 |
| ETR $P_2$ | 1 | 10 | 50 | 0.1 | 1 | 5.0 |
| ~~Arithmetic mean~~ | ~~1~~ | ~~5.05~~ | ~~25.025~~ | ~~5.05~~ | ~~1~~ | ~~2.75~~ |

# Summarizing Performance

## Geometric Mean

- Independent of reference
- No physical meaning

$$Geometric\ Mean = GM = \sqrt[n]{\prod_{i=1}^{n} ExecutionTimeRatio_i}$$

# Summarizing Performance

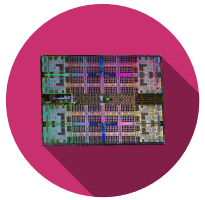
## Geometric Mean - Example

$$GM = \sqrt[n]{\prod_{i=1}^{n} ExecutionTimeRatio_i}$$

| | CPU A | CPU B | CPU C |
|---|---|---|---|
| Program $P_1$ (sec) | 1 | 10 | 20 |
| Program $P_2$ (sec) | 1000 | 100 | 20 |

| | Normalized to A | | | Normalized to B | | |
|---|---|---|---|---|---|---|
| | CPU A | CPU B | CPU C | CPU A | CPU B | CPU C |
| ETR $P_1$ | 1 | 0.1 | 0.05 | 10 | 1 | 0.5 |
| ETR $P_2$ | 1 | 10 | 50 | 0.1 | 1 | 5.0 |
| Geometric mean | 1 | 1 | **1.58** | 1 | 1 | **1.58** |

# CPU Performance Equation



$$T = IC * CPI * CT = \frac{IC * CPI}{f}$$

- $T = \textit{Execution time}$
- $IC = N_{instr}$ = # instructions executed (**I**nstruction **C**ount)
- $CPI$ = **C**ycles **P**er **I**nstruction
- $CT = t_{cycle}$ = **C**ycle **T**ime = duration of clock cycle
- $f$ = clock frequency = $\frac{1}{t_{cycle}}$

# CPU Performance Equation

## Improving Performance

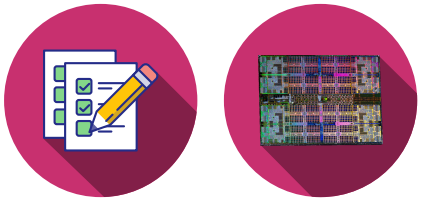
$$T = IC * CPI * CT = \frac{IC * CPI}{f}$$

- Factors are interdependent => Trade-offs
  - Cost
  - Power
  - Performance

| Reduce | How | Side effect |
|---|---|---|
| $IC$ / $N_{instr}$ | Increase capability / complexity of instructions | $CPI$ / $t_{cycle}$ increases |
| $CPI$ | Simple instructions required fewer cycles | $IC$ / $N_{instr}$ increases |
| $CT$ / $t_{cycle}$ | Less work per cycle | $CPI$ increases |

# CPU Performance Equation

## Calculating Execution Time

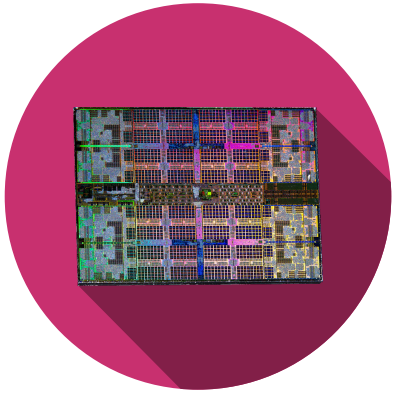


A program executes 5 Million instruction with a given instruction mix:

| Instruction | Frequency | $CPI_{instr}$ |
|---|---|---|
| ALU | 50% | 3 |
| Load | 20% | 5 |
| Store | 10% | 4 |
| Branch | 20% | 3 |

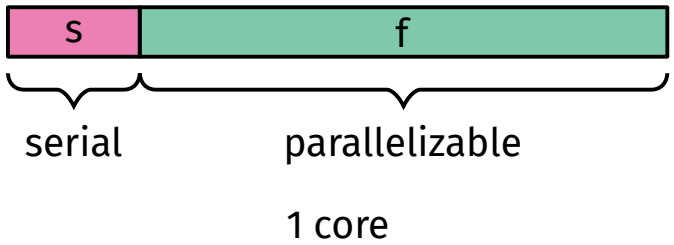
- The CPI has a frequency of 2GHz

1. What is the execution time?



# Amdahls Law

# Amdahl's Law



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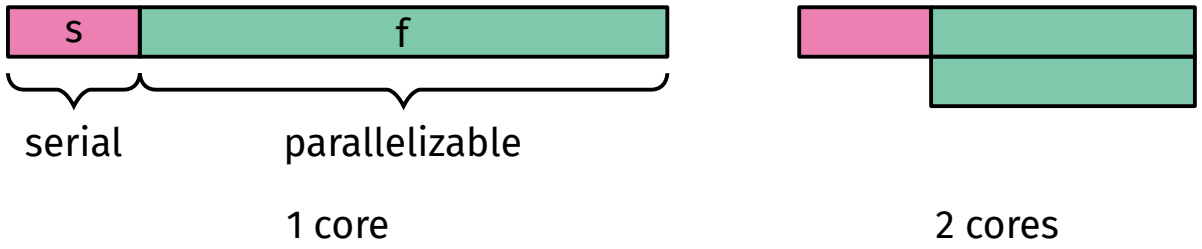

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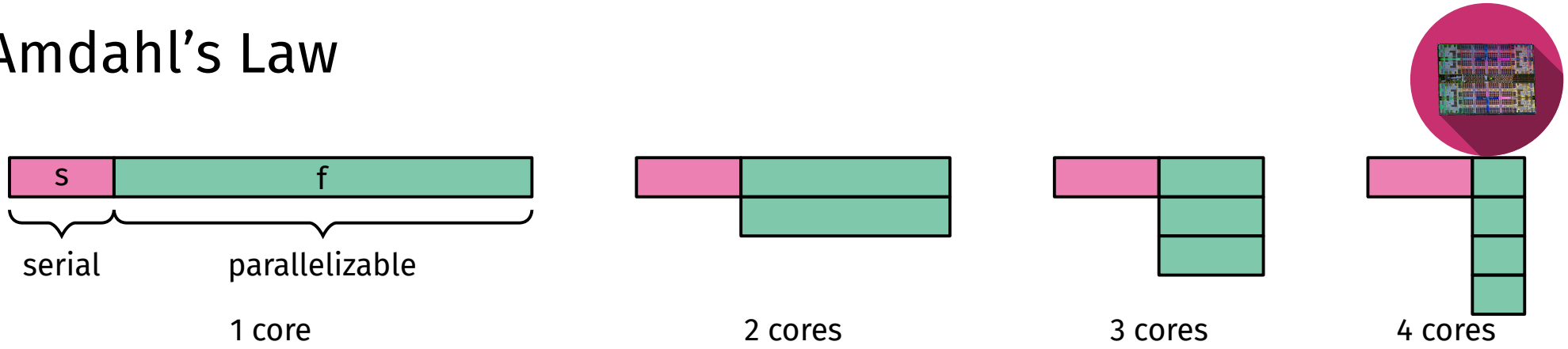

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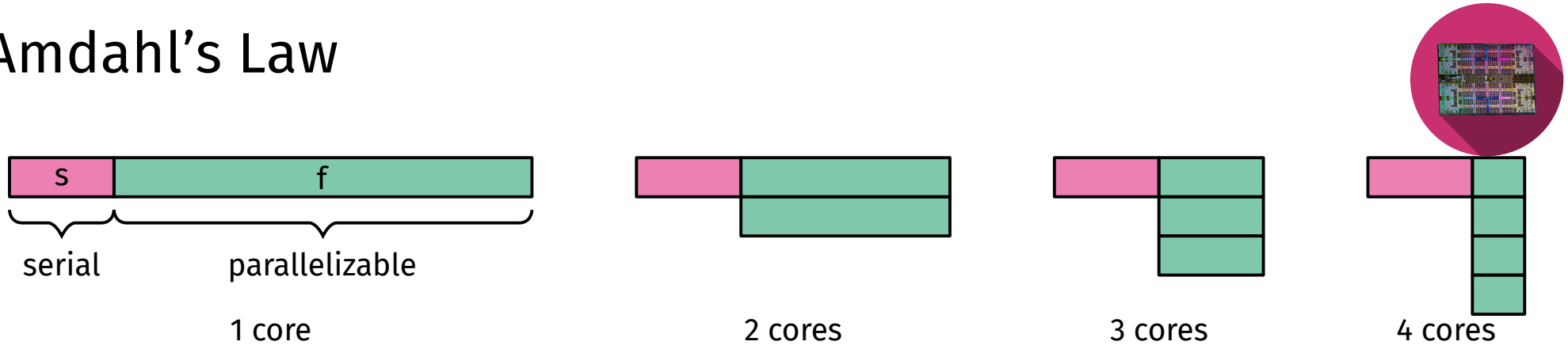
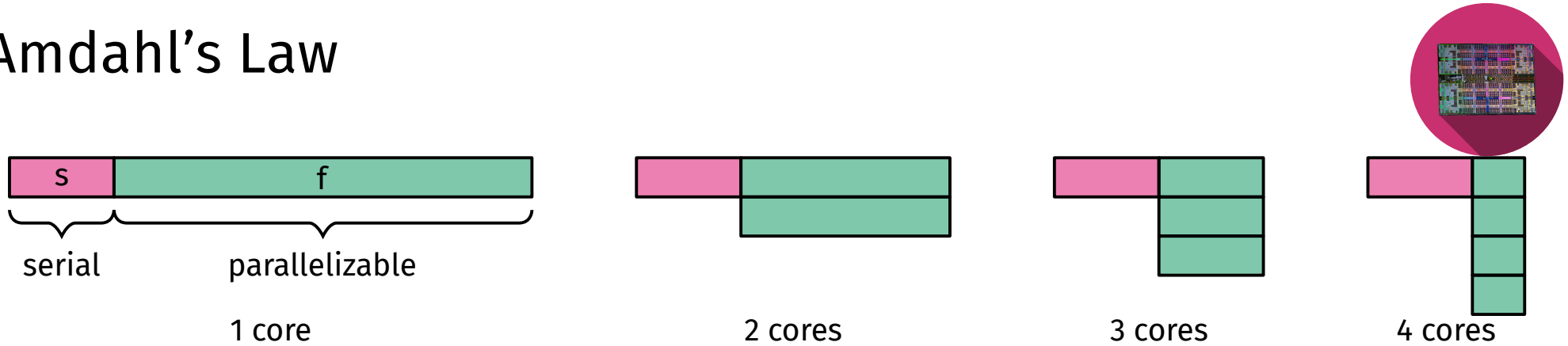
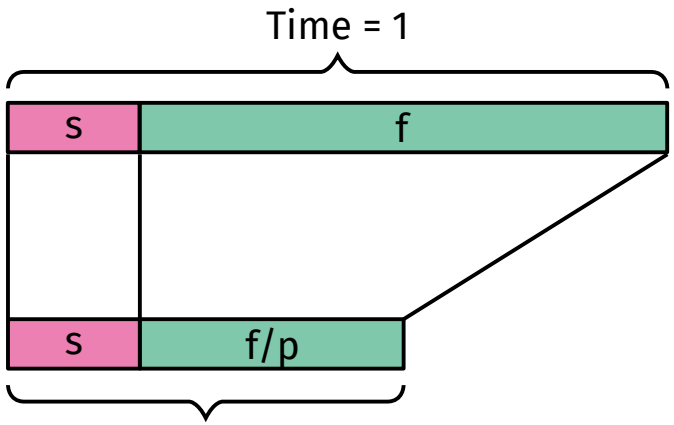



If 50% of the execution time is sequential, the maximum speedup is 2, no matter how many cores you use.
- Gene Amdahl -

# Amdahl's Law




$$Time = s + \frac{f}{p}$$

$$S = \frac{1}{s + \frac{f}{p}} = \frac{1}{1 - f + \frac{f}{p}}$$

$S$ = Speedup
$s = 1 - f = serial\ fraction$
$f = 1 - s$ = parallel fraction
$p = \#\ processor\ cores$

# Amdahl’s Law

## Speedup for Different Parallel Fractions

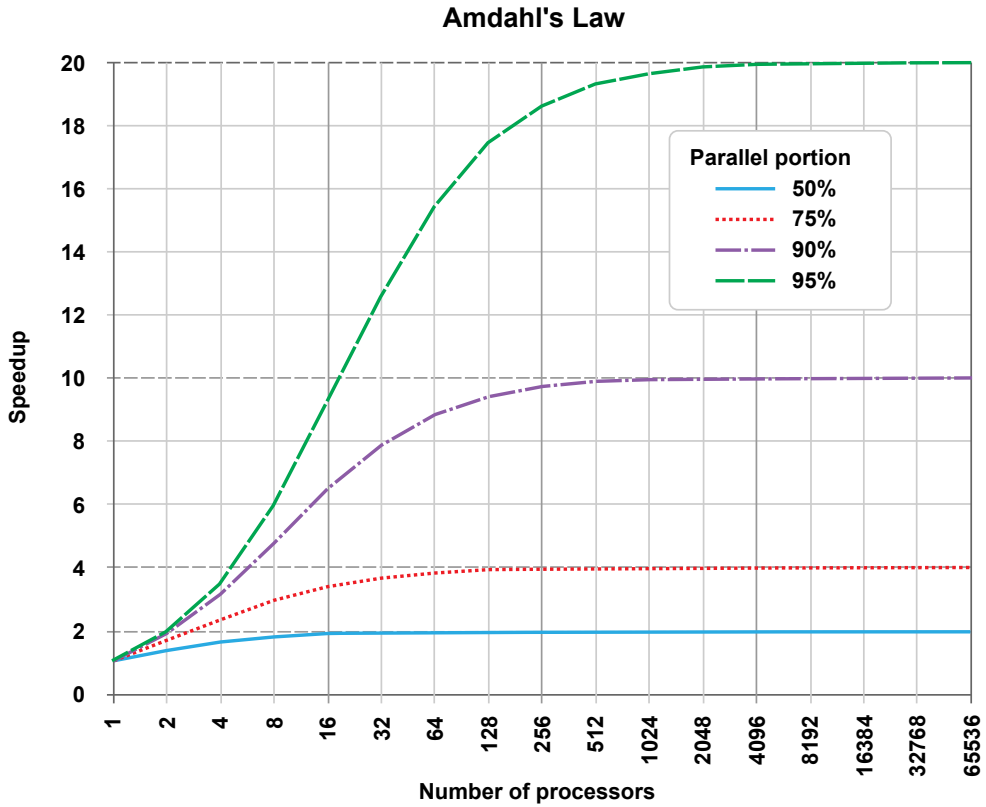
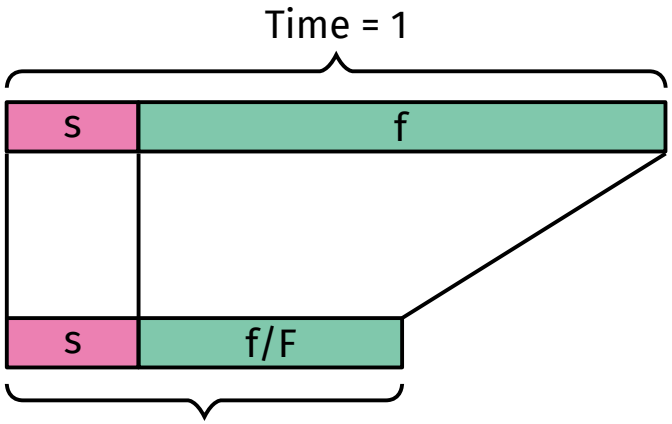



*Image Credit: Wikipedia*

# Amdahl's Law

## Generalization of Amdahl's Law

- Suppose we can improve fraction f by factor F
- What will be overall improvement?



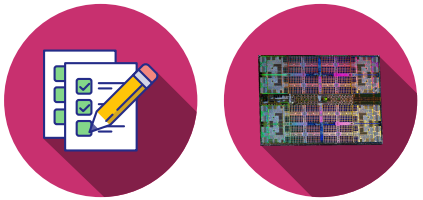

$$Time = s + \frac{f}{F}$$

$$S = \frac{1}{s + \frac{f}{F}} = \frac{1}{1 - f + \frac{f}{F}}$$

$S$ = Speedup
$s = 1 - f$ = fraction we cannot improve
$f = 1 - s$ = fraction we can improve
$F = improvement\ factor$

# Amdahl's Law

## Example



*FP instruction improved to run 2x as fast, but only 10% of all executed instructions are FP.*

- *What will be the overall speedup?*

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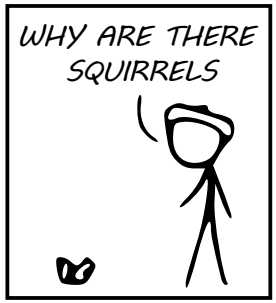
# QUESTIONS

CAN BE ASKED BY ANYONE ANYTIME

WHY ARE THERE MIRRORS ABOVE BEDS
WHY DO I SAY UH
WHY IS SEA SALT BETTER
WHY ARE THERE TREES IN THE MIDDLE OF FIELDS
WHY IS THERE NOT A POKEMON MMO
WHY IS THERE LAUGHING IN TV SHOWS
WHY ARE THERE DOORS ON THE FREEWAY
WHY ARE THERE SO MANY SVCHOST.EXE RUNNING
WHY AREN'T ANY COUNTRIES IN ANTARCTICA
WHY ARE THERE SCARY SOUNDS IN MINECRAFT
WHY IS THERE KICKING IN MY STOMACH
WHY ARE THERE TWO SLASHES AFTER HTTP
WHY ARE THERE CELEBRITIES
WHY DO SNAKES EXIST
WHY DO OYSTERS HAVE PEARLS
WHY ARE DUCKS CALLED DUCKS
WHY DO THEY CALL IT THE CLAP
WHY ARE KYLE AND CARTMAN FRIENDS
WHY IS THERE AN ARROW ON AANG'S HEAD
WHY ARE TEXT MESSAGES BLUE
WHY ARE THERE MUSTACHES ON CLOTHES
WHY WUBA LUBBA DUB DUB MEANING
WHY IS THERE A WHALE AND A POT FALLING
WHY ARE THERE SO MANY BIRDS IN SWISS
WHY IS THERE SO LITTLE RAIN IN WALLIS
WHY IS WALLIS WEATHER FORECAST ALWAYS WRONG
WHY ARE THERE MALE AND FEMALE BIKES
WHY ARE THERE BRIDESMAIDS
WHY DO DYING PEOPLE REACH UP
HOW FAST IS LIGHTSPEED
WHY ARE OLD KLINGONS DIFFERENT

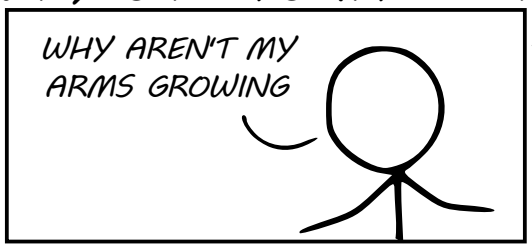



WHY AREN'T THERE DINOSAUR GHOSTS
WHY IS THERE HELL IF G

WHY IS THERE CAFFEINE IN MY SHAMPOO
WHY HAVE DINOSAURS NO FUR
WHY DO IGUANAS DIE
WHY ARE SWISS AFRAID OF DRAGONS
WHY AREN'T ECONOMISTS RICH
WHY DO AMERICANS CALL IT SOCCER
WHY ARE MY EARS RINGING
WHY IS 42 THE ANSWER TO EVERYTHING
WHY CAN'T NOBODY ELSE LIFT THORS HAMMER
WHY IS MARVIN ALWAYS SO SAD
WHY IS THERE LAVA
WHY IS THERE A SWARM OF ANTS
WHY IS THERE PILEGRIM
WHY ARE THERE ANTS IN MY LAPTOP
WHY IS EARTH TILTED
WHY IS SPACE BLACK
WHY IS OUTER SPACE SO COLD
WHY ARE THERE PYRAMIDS ON THE MOON
WHY IS NASA SHUTTING DOWN
WHY ARE THERE TINY SPIDERS IN MY HOUSE
WHY DO SPIDERS COME INSIDE
WHY ARE THERE HUGE SPIDERS IN MY HOUSE
WHY ARE THERE LOTS OF SPIDERS IN MY HOUSE
WHY ARE THERE SPIDERS IN MY ROOM
WHY ARE THERE SO MANY SPIDERS IN MY ROOM
WHY DO SPYDER BITES ITCH
WHY IS DYING SO SCARY
WHY IS THERE NO GPS IN LAPTOPS
WHY DO KNEES CLICK
WHY ARE THERE FEMALE

WHY 2*B | 2*B

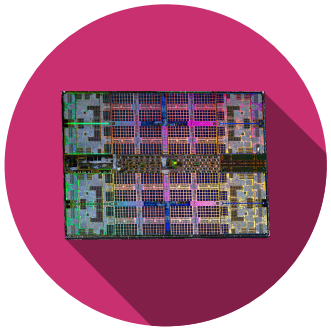
WHY IS THERE A LINE THROUGH HTTPS
WHY IS THERE A RED LINE THROUGH HTTPS ON TWITTER
WHY IS HTTPS IMPORTANT
WHY IS YKK ON ALL ZIPPERS




WHY ARE THERE WEEKS
WHY DO I FEEL DIZZY
WHY ARE THERE SO MANY CROWS IN ROCHESTER
WHY IS TO BE OR NOT TO BE FUNNY
WHY DO CHILDREN GET CANCER
WHY IS POSEIDON ANGRY WITH ODYSSEUS
WHY IS THERE ICE IN SPACE
WHY ARE DOGS AFRAID OF FIRE
WHY IS THERE NO KING IN EN
WHY IS THERE AN OWL IN MY BACKYARD
WHY IS THERE AN OWL OUTSIDE MY WINDOW
WHY IS THERE AN OWL ON THE DOLLAR BILL
WHY DO OWLS ATTACK PEOPLE
WHY ARE FPGA's EVERYWHERE
WHY ARE THERE HELICOPTERS CIRCLING MY HOUSE
WHY ARE THERE GODS
WHY ARE THERE TWO SPOCKS
WHY ARE MY BOOBS ITCHY
WHY ARE CIGARETTES LEGAL
WHY ARE THERE DUCKS IN MY POOL
WHY IS JESUS WHITE
WHY IS THERE LIQUID IN MY EAR
WHY DO Q TIPS FEEL GOOD
WHY DO PEOPLE DIE
WHY IS LIFE SO
WHAT IS https://xkcd.com/1256/
WHY DO THEY SAY T-MINUS
WHY ARE THERE OBELISKS
WHY ARE WRESTLERS ALWAYS WET
WHY ARE OCEANS BECOMMING MORE ACIDIC

Hes·so VALAIS WALLIS

Haute Ecole d'Ingénierie
Hochschule für Ingenieurwissenschaften



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